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Indirect reciprocity among imperfect individuals

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Abstract

Reciprocal cooperation occurs when the overall benefits of receiving help exceed the costs of donating help (Q. Rev. Biol. 46 (197) 35). That is, individuals in good condition—for whom the pertinent costs are relatively small; donate help in order to secure reciprocity in their hour of need—when the benefits of receiving a donation are large. Consequently, reciprocity occurs among individuals who occasionally need help. In particular, such individuals will be unable to help others, no matter how deserving, when in need of help themselves—*involuntary defection*. This paper deals with the effects of involuntary defection in the context of a specific model of indirect reciprocity (i.e. reciprocal altruism that is directed toward all the cooperative members of the community) due to Nowak and Sigmund (J. Theor. Biol.194 (1998b) 561: Sections 2–4). In that model, the authors formulate the decision rules for conditional cooperation in the context of indirect reciprocity, and demonstrate that these decision rules can account for a long-term persistence of cooperation.

Here we show that addition of involuntary defection to the decision rules formulated by Nowak and Sigmund results in indirect reciprocity that is evolutionary stable under appropriate conditions. Moreover, for a wide range of parameter values, evolutionary stability of cooperation requires a mixture of conditional- and unconditional-altruist behaviors. To recollect, unconditional altruist strategy can be viewed as conditional altruist strategy *sans* the ability to decide when the help-soliciting individual should be refused help. That is, given involuntary defection, stability of cooperation requires an occasional forgiveness, if only by default, of a failure to donate help.

Thus, we see that evolutionary stable indirect reciprocity does not require perfection in either the ability to assess the merits of the help-soliciting individuals, or the ability to donate help when it is merited. On the contrary, we are forced to conclude that reciprocity, at least in the current case, is stable only among imperfect individuals.
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1. Introduction

Altruistic behavior is commonly attributed to inclusive fitness, or reciprocity (Krebs and Davies, 1993). The idea of reciprocal altruism, albeit one motivated by conscious calculation, was first proposed by Darwin: "... as the reasoning powers and foresight become improved, each man would soon learn from experience that if he aided his fellow men, he would commonly receive aid in return" (Darwin, 1871). The idea of reciprocal altruism in its modern form, i.e. as an unconscious heritable trait maintained by Darwinian fitness advantages, was first advanced by Williams (1966), and given rigor by Trivers (1971).

Trivers used game theoretical methods to show that an individual may help an unrelated conspecific provided: (i) the cost to the donor is less than the benefit to the recipient and (ii) the favor is likely to be returned at a later date. That is, reciprocal altruism (cooperation) works only if individuals receive more than they donate. In particular, cooperators must protect themselves from exploitation by individuals who solicit favors, but do not reciprocate—*defectors*.

Trivers, with the aid of W.D. Hamilton (Trivers, 1971, p. 39), resolved this issue by postulating that cooperation is conditional, i.e. a conditional cooperator will keep helping an unrelated individual, unless the later refuses to reciprocate. These theoretical results soon gained empirical support from observations detailing pairwise exchanges of favors—*direct reciprocity* (Fisher, 1980; Seyfarth and Cheney, 1984;

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Wilkinson, 1984). The subject of direct reciprocity was further developed by Axelrod and co-workers (Axelrod and Hamilton, 1981) resulting in the formulation of *Tit-for-Tat* strategy: a Tit-for-Tat player will punish defection by refusing help in turn, but otherwise will cooperate.

In his 1971 paper, Trivers also raises the possibility of a "generalized altruism" where the return is directed towards a third party. That is: "In a system of strong multiparty interactions, it is possible that in some situations individuals are selected to demonstrate generalized altruistic tendencies."

Alexander (1979, 1987) extended the concept of generalized altruism under the name of *indirect reciprocity*. In indirect reciprocity, cooperation is channeled towards the cooperative members of the community. To wit, help is donated if the recipient is likely to help others (which is decided on the basis of experience, i.e. according to whether the potential recipient has helped others in the past).

The idea of indirect reciprocity was given an analytical form by Nowak and Sigmund (1998a, b). In their 1998a paper, Nowak and Sigmund have advanced the idea of indirect reciprocity maintained through *image scoring*. The main points of this approach are as follows.

- Each individual has an image score. Each time an individual donates help its score increases by a unit. If an individual refuses a request for help, its image score decreases by a unit.
- When individuals with whom they did not have previous direct interactions solicit help, conditional cooperators (*discriminating altruists*) donate help only if the image score of the applicant is greater or equal to their own.

In their 1998b paper, Nowak and Sigmund have further refined the idea of indirect reciprocity. In particular, they show that indirect reciprocity can occur even if discriminators only remember the help-soliciting individual's score for a single interaction, and only refuse to help individuals who failed to donate help to others. That is, indirect reciprocity can be implemented by the indirect analog of the dyadic Tit-for-Tat strategy.

Nowak and Sigmund's papers led to a renewal of interest in indirect reciprocity, prompting intense research efforts involving both theoretical and experimental approaches (Lotem et al., 1999; Wedekind and Milinski, 2000; Milinski et al., 2001; Leimar and Hammerstein, 2001; Milinski et al., 2002a,b; Wedekind and Braithwaite, 2002).

However, indirect reciprocity in Nowak and Sigmund (1998b) model(s) is persistent (lasts for long periods of time under appropriate conditions) rather than evolutionary stable. Here we show that this result is not an inherent attribute of the authors' discriminating altruist strategy, but is rather an artifact of one of their model's implicit assumptions.

- In their initial analytical formulation of a very complex situation, the authors focused on the rules governing the assessment of the help-soliciting individuals by conditional cooperators, while assuming—a standard feature of reciprocity modeling prior to Lotem et al. (1999), that a generic player is able to donate help whenever required by the pertinent decision rules.

Clearly, this is an approximation—in practice individuals who are always able to donate help do not need help from others. Thus, in modeling reciprocity, we deal with individuals who donate help when their costs are small, in order to secure reciprocity in their hour of need—when the benefits of receiving a donation are large. In particular, such individuals will be unable to help others, no matter how deserving, when in need of help themselves: *involuntary-* or *phenotypic-defection* (cf. Lotem et al., 1999; Fishman et al., 2001; Sherratt and Roberts, 2001).

Here we show that involuntary defection stabilizes indirect reciprocity. This result has a straightforward and, in retrospect, an almost self-evident explanation. If involuntary defection is neglected, the interactions between unconditional and conditional altruists are symmetric. Thus, these two strategic types have identical fitness whenever they constitute an entire population. Consequently, unconditional altruists can spread through such a population by genetic drift. Finally, (genetic) defectors can invade a population containing a substantial fraction of unconditional altruists. However, in real-world populations, this symmetry does not exist: conditional altruists punish unconditional altruists for involuntary defection but not vice versa.

Finally, we were able to address another issue raised by Nowak and Sigmund (1998b, Section 8): the practical/desirable limits to the individual's ability to monitor and remember interactions among others. We see that under appropriate conditions, evolutionary stable cooperation requires a mixture of discriminating and indiscriminate altruist behaviors. To recollect, unconditional altruists can be viewed as conditional altruists who have lost the discriminating facility (Selten and Hammerstein, 1984). Thus, we see that a degree of imperfection in the ability to keep image scores is essential for the maintenance of indirect reciprocity.

Although this last result seems to be counterintuitive, it has a simple interpretation. In the presence of involuntary defection, conditional altruists' refusal to help individuals who have been observed to defect in the past can actually decrease the overall level of cooperation. That is, given a complete fidelity in image score

keeping, an individual penalizing another for an involuntary defection, incurs a penalty for itself, and the individual responsible is penalized in turn—a chain reaction that can lead to the collapse of cooperation. As we show below, there is a precise degree of fidelity in image score keeping that dampens such a chain reaction while still preventing an invasion by (voluntary) defectors.

This paper is organized as follows: The current model is the game developed in Sections 2 through 4 of Nowak and Sigmund (1998b) with the addition of involuntary defection. However, as the resultant model differs from the original in non-obvious ways, we detail its construction in Section 2. In Section 3, we analyse the resulting game and discuss its salient futures. To facilitate presentation, we confine the technical details of the analysis to the appendix.

2. Formulation of the model

It is assumed that each generation experiences $N \geqslant 2$ rounds of interactions. In each round, a player both solicits and is solicited for help. As Nowak and Sigmund show (1998b, Section 6), this idealization simplifies the calculations without changing the fundamental properties of the game. Following Nowak and Sigmund (1998b), we denote the benefits of the altruistic act to the recipient by b, the corresponding costs to the donor by $c > 0$, and assume $b > c$. Finally, we define the probability that an average player is able (as distinct from willing) to donate help on request by $0 < r \leqslant 1$.

We start by defining the pure strategies of the game:

Indiscriminate altruists (**IA**). Donate help whenever it is solicited, provided they can afford such donation at the time (i.e. there is a probability $1 - r$ that an applicant is denied help).

Unconditional defectors (**UD**). Solicit but do not donate help.

Discriminating altruists (**DA**). Refuse help to defectors in the previous round, and are constrained by their own ability to make a donation.

We denote the frequencies of the **IA**, **UD**, and **DA** players by x_1, x_2, and x_3 respectively, That is, the pure strategies **IA**, **UD**, and **DA** are represented by the successive elements: $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$; of the standard basis of $\mathbf{R}^3$.

Remark. In Section 8 of their paper, Nowak and Sigmund introduce the concept of *incomplete information*. To wit, they specify a probability $0 < q < 1$ that a **DA** player knows the score of the help-soliciting individual. However, they also postulate that, in the absence of the relevant information, **DA** players act as **IA** players. Thus, as we shall see below, the possibility of incomplete information is represented in the current model as a mixed strategy, $\mathbf{IA} \oplus \mathbf{DA}$.

Now we can consider the payoffs: An average indiscriminate altruist is always willing to donate help, but is only able to do so with probability r. In particular, this affects their interactions with discriminating altruists. Specifically, in the second and all subsequent rounds **DA** players treat a fraction $1 - r$ of **IA** players as "undeserving". Thus, **IA** players receive help from **DA** players with probability r in the first round, and probability r^2 in all subsequent rounds. Interactions among **IA** players are symmetric, and they never receive help from **UD** players. Thus, the payoff to a generic **IA** player is given by

$$P_1(\mathbf{x}) = Nr[(b-c)x_1 - cx_2] + r[N(rb-c) + (1-r)b]x_3. \tag{1}$$

UD players never help others and receive help from **IA** players with probability r, and help from **DA** players with probability r in the first round, and probability zero in subsequent rounds. Thus, **UD** players' payoffs are given by

$$P_2(\mathbf{x}) = rb(Nx_1 + x_3). \tag{2}$$

DA players receive help from **IA** players with probability r. They help **IA** players with probability r in the first round, and probability r^2 in all subsequent rounds. **DA** players help **UD** players with probability r in the first round and do not help **UD** players in all subsequent rounds.

To formulate a functional expression for reciprocity among **DA** players we proceed as follows. Let us define the probability that a generic **DA** player donated help in the n-th round by $g_n(\mathbf{x})$. Hence, the fraction of individuals who were helpful in the n-th round, and therefore deserve help, is given by $rx_1 + g_n(\mathbf{x})x_3$. Thus, given involuntary defection, the probability that a generic **DA** player donates help in the $(n+1)$-th round, $g_{n+1}(\mathbf{x})$, is given by

$$g_{n+1}(\mathbf{x}) = r[rx_1 + g_n(\mathbf{x})x_3]. \tag{3a}$$

Hence, since $g_1(\mathbf{x}) = r$, we see that

$$g_n(\mathbf{x}) = r\left[(rx_1)\sum_{k=0}^{n-2}(rx_3)^k + (rx_3)^{n-1}\right]. \tag{3b}$$

And, therefore,

$$\sum_{n=1}^{N} g_n(\mathbf{x}) = r[1 + \phi(\mathbf{x})], \tag{4}$$

where

$$\phi(\mathbf{x}) = rx_1\sum_{k=0}^{N-2}(N-k-1)(rx_3)^k + \sum_{k=1}^{N-1}(rx_3)^k.$$

Consequently, the total payoff to a generic **DA** player is given by

$$P_3(\mathbf{x}) = r[N(b-rc) - (1-r)c]x_1 - rcx_2 + r[1+\phi(\mathbf{x})](b-c)x_3. \quad (5)$$

Notice that the interactions between **DA** and **IA** strategists are not symmetric. **DA** players receive more help than they donate. As we shall see below, this fact plays a crucial role in determining the properties of the current model.

3. Properties of the game

The payoffs in evolutionary games represent fitness, and evolutionary stability depends on the *relative fitness* of the game's strategies. Thus, we can subtract a given term from all payoffs, and can also divide all payoffs by a non-zero constant, without changing the set of the game's *evolutionary stable strategies* (ESSs). Such transformations have the virtue of both simplifying the algebraic manipulations involved in evolutionary stability analysis, and (ideally) emphasizing the principal features of the model.

Here we subtract $P_2(\mathbf{x}) - rc$ from all the payoffs, and divide the result by the constant rb to obtain

$$\pi_1(\mathbf{x}) = (N-1)(rx_3 - \gamma),$$
$$\pi_2(\mathbf{x}) = \gamma,$$
$$\pi_3(\mathbf{x}) = \phi(\mathbf{x})(1-\gamma)x_3 - (N-1)r\gamma x_1, \quad (6)$$

where $\gamma = c/b$ is the *relative* cost of reciprocity. Note: since $b > c > 0$, $0 < \gamma < 1$.

In the appendix, we show that system (6) has three ESS solutions:

- **UD** is an ESS for all parameter values.
- **DA** is an ESS whenever

$$\gamma < \gamma_1 < \gamma_2 \quad \text{and} \quad \rho_1 < r < \rho_2$$
$$\text{or } \gamma_1 < \gamma < \gamma_2 \quad \text{and} \quad \rho_1 < r < 1. \quad (7)$$

Here $\gamma_1 = (N-2)/N$ and $\gamma_2 = (N-1)/N$. The threshold $0 < \rho_1 < 1$ is the unique solution of the equation $(1-\gamma)\sum_{k=1}^{N-1} r^k = \gamma$, if $\gamma < \gamma_2$. Finally, when $\gamma < \gamma_1, 0 < \rho_2 < 1$ is the unique solution, of the equation $(1-\gamma)\sum_{k=0}^{N-2}(N-k-1)r^k = N-1$. In the appendix, Eqns. (A.10), we show that $\rho_1 < \rho_2$ if both are defined.

We depict the variation of ρ_1 and ρ_2 with γ and N in Fig. 1.

- Finally, if $N > 2$, $\gamma < \gamma_1$, and $\rho_2 < r < 1$, there is an evolutionary stable, mixed strategy solution **IA** $\oplus$ **DA** for which the frequency of **DA** behavior is η, where η is prescribed by the equation

$$(1-\gamma)\eta \sum_{k=1}^{N-2} (N-k-1)(r\eta)^k = (N-1)\gamma. \quad (8)$$

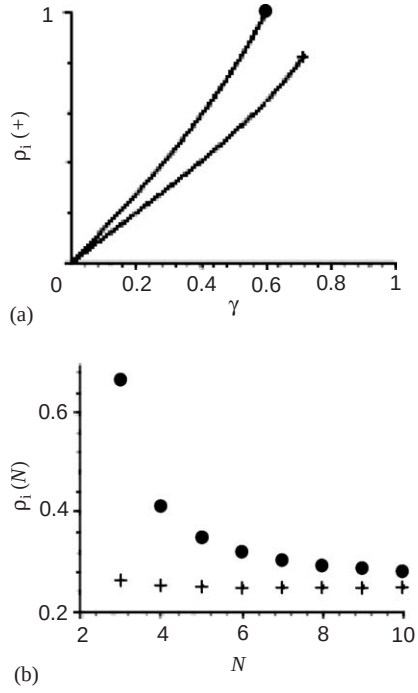


Fig. 1. (a) $\rho_1(+)$ and $\rho_2(\bullet)$ vs. γ, for $N = 5$. (b) ρ_1 and ρ_2 vs. N, for $\gamma = 0.25$.

The variation of η with γ, N, and r is presented in Fig. 2.

We see that, even for relatively low values of N,[1] a pure strategy solution **DA** is an ESS only over a narrow range of parameter values. That is, indirect reciprocity is more likely to be implemented via the mixed strategy **IA** $\oplus$ **DA**.

As discussed in Section 2, we may view the mixed strategy **IA** $\oplus$ **DA** as representing a conditional cooperation strategy for situations where individuals must operate on the basis of incomplete information. Alternatively, we may view **IA** $\oplus$ **DA** as a population-level polymorphism wherein the frequency of **DA** strategists is η, and the frequency of the **IA** strategists is $1-\eta$. In any case, if $N > 2$, $\gamma < \gamma_1$, and $\rho_2 < r < 1$, cooperation is stable only when the probability of a conditional response (i.e. a response to a request for help depends on the image score of the help-soliciting individual) is equal to η.

4. Discussion

An individual may help an unrelated conspecific whenever: (i) the cost to the donor is less than the

[1] It can be shown (appendix: eqns. (A.10) that ρ_1 and ρ_2 converge to γ when N tends to infinity. Thus, at the limit $N \to \infty$, system (6) has very simple description. If $r < \gamma$, **UD** is the only ESS. If $\gamma < r < 1$, **IA** $\oplus$ **DA** is also an ESS. However, the original model (Nowak and Sigmund, 1998b) restricts N to low values so as to reduce the probability of repeated dyadic encounters (direct reciprocity).

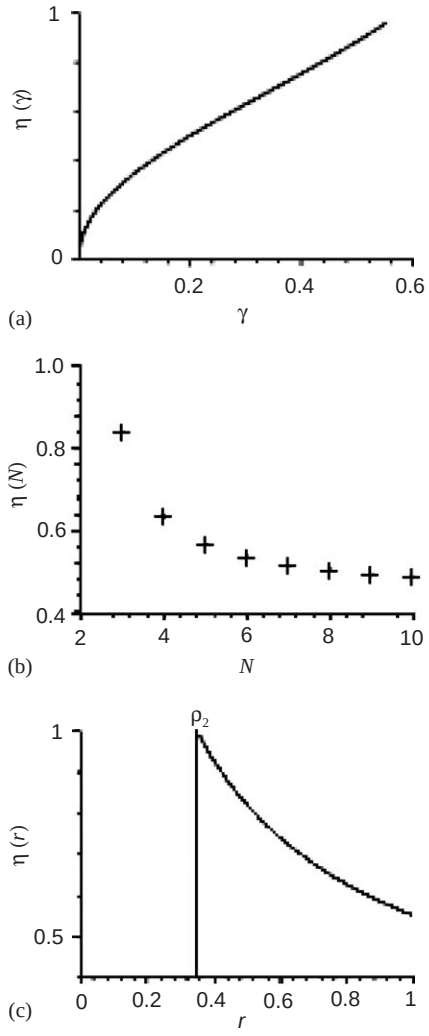


Fig. 2. (a) η vs. γ for $N = 5$ and $r = 0.95$. (b) η vs. N for $\gamma = 0.25$ and $r = 0.95$. (c) η vs. r for $\gamma = 0.25$ and $N = 5$.

benefit to the recipient, and (ii) the favor is likely to be returned at a latter date (Trivers, 1971). That is, individuals in good condition—for whom the associated fitness costs are relatively small—donate help to others in order to secure reciprocity in their hour of need—when the fitness benefits of receiving a donation are large. Consequently, reciprocity should only occur among individuals who occasionally need help. However, such individuals will not be able to help others, no matter how deserving, when in need of help themselves.

The possibility that individuals may be unable, as distinct from unwilling, to donate help upon request was first addressed by Lotem and co-workers (Lotem et al., 1999; Fishman et al., 2001), who named this phenomena *phenotypic defection*. However, Lotem et al. were primarily interested in the quality differences among individuals, and developed the subject of phenotypic defection in the direction of the possible uses of helpful behavior as a signal of quality (Lotem et al., 2002). Specifically, Lotem and co-workers divide modeled populations into high-quality individuals—who are always able to donate help; and low-quality phenotypic defectors who are unable to donate help. Hence, I feel that the term *phenotypic defection* should be reserved for models focusing on individual differences, and propose the term *involuntary defection* for describing the average individual's occasional inability to donate help.

This paper deals with the issue of involuntary defection in the context of a specific model of indirect reciprocity due to Nowak and Sigmund (1998b, Sections 2–4). In that model, the authors formulate the decision rules for conditional cooperation in the context of indirect reciprocity, and demonstrate that these decision rules can account for a long-term persistence of cooperation.

Here we show that addition of involuntary defection to the decision rules formulated by Nowak and Sigmund results in indirect reciprocity that is evolutionary stable under a wide range of parameter regimes. This result has a straightforward explanation. If involuntary defection is neglected, the interactions between unconditional and conditional altruists are symmetric. Thus, these two strategic types have identical fitness whenever they constitute an entire population. Consequently, unconditional altruists can spread through such a population by genetic drift. And a population containing a substantial fraction of indiscriminate altruists can be invaded by (voluntary) defectors. However, in real-world populations this symmetry does not exist: conditional altruists punish unconditional altruists for involuntary defection but not vice versa.

Finally, we see that under appropriate conditions, evolutionary stable cooperation requires a mixture of discriminating and indiscriminate altruist behaviors. To recollect, unconditional altruist strategy can be viewed as conditional altruist strategy *sans* the ability to assess the merits of the help-soliciting individuals—resulting in indiscriminate donation of help (Selten and Hammerstein, 1984). Thus, our results indicate that stable cooperation requires an occasional "forgiveness" of defection. In retrospect, this result should not be surprising as penalizing involuntary defection is counterproductive.

To summarize. Our results indicate that evolutionary stable indirect reciprocity does not require perfection in either the ability to assess the merits of the help-soliciting individuals, or the ability to donate help when it is merited. On the contrary, we are forced to conclude that reciprocity, at least in the current case, is stable only among imperfect individuals.

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Appendix. Methods and notation

Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be the *standard basis* of $\mathbf{R}^3$. And let us define the strategy set for the game by

$$\mathbf{X} = \left\{ \sum_{j=1}^{3} x_j \mathbf{e}_j \,\middle|\, x_1, x_2, x_3 \geqslant 0 \text{ and } x_1 + x_2 + x_3 = 1 \right\}. \tag{A.1}$$

When the frequencies of the pure strategies in the population are given by $\mathbf{p} \in \mathbf{X}$, the payoff for playing an arbitrary strategy $\mathbf{x} \in \mathbf{X}$ is given by

$$v(\mathbf{x}, \mathbf{p}) = \mathbf{x}^{\mathrm{t}} \circ \begin{pmatrix} \pi_1(\mathbf{p}) \\ \pi_2(\mathbf{p}) \\ \pi_3(\mathbf{p}) \end{pmatrix}. \tag{A.2}$$

It can be shown (cf. Hofbauer and Sigmund, 1998, p. 65) that $\mathbf{x}^* \in \mathbf{X}$ is an ESS of system (6) *if*, *and only if*, (*iff*) there exists $0 < \varepsilon^* \ll 1$ such that for every strategy $\mathbf{x} \neq \mathbf{x}^*$ and every $0 < \varepsilon \leqslant \varepsilon^*$

$$v(\mathbf{x}^* - \mathbf{x}, \mathbf{x}_\varepsilon) > 0$$
$$\text{where } \mathbf{x}_\varepsilon = \mathbf{x}^* + \varepsilon(\mathbf{x} - \mathbf{x}^*). \tag{A.3}$$

For $k \geqslant 0$ let us define

$$\xi_k(\mathbf{x}^*, \mathbf{x}) = \frac{\partial^k v}{\partial \varepsilon^k}(\mathbf{x}^* - \mathbf{x}, \mathbf{x}_\varepsilon)\bigg|_{\varepsilon=0}. \tag{A.4a}$$

Thus,

$$v(\mathbf{x}^* - \mathbf{x}, \mathbf{x}_\varepsilon) = \sum_{k=0}^{\infty} \xi_k(\mathbf{x}^*, \mathbf{x}) \varepsilon^k. \tag{A.4b}$$

Hence, $\mathbf{x}^* \in \mathbf{X}$ is an ESS of system (6) *iff* for every strategy $\mathbf{x} \neq \mathbf{x}^*$ there exist $n \geqslant 0$ such that

$$\xi_n(\mathbf{x}^*, \mathbf{x}) > 0 \quad \text{and} \quad 0 \leqslant k < n \Rightarrow \xi_k(\mathbf{x}^*, \mathbf{x}) = 0. \tag{A.5}$$

Remark. We cannot test every strategy in $\mathbf{X}$ for being an ESS. However, it is easy to show that a strategy is an ESS of a game *only if* it is a stationary solution of that game's replicator equations. Thus, we analyse system (6) as a two-step process.

(i) We find all the stationary solutions of the associated replicator equations.
(ii) We test these potential ESS solutions with the ESS criterion (A.5).

System (6) has six potential ESS solutions.

- The three pure strategy solutions: $\mathbf{x}_1 = \mathbf{e}_1 \leftrightarrow \mathbf{IA}$, $\mathbf{x}_2 = \mathbf{e}_2 \leftrightarrow \mathbf{UD}$, and $\mathbf{x}_3 = \mathbf{e}_3 \leftrightarrow \mathbf{DA}$.
- A mixed strategy $\mathbf{x}_4 = (1-\eta)\mathbf{e}_1 + \eta\mathbf{e}_3 \leftrightarrow \mathbf{IA} \oplus \mathbf{DA}$, where (under appropriate conditions on the parameters) $0 < \eta < 1$ is the unique root of the equation

$$H(\eta) \equiv S(\eta) - (N-1)\gamma = 0,$$
$$\text{where } S(\eta) \equiv (1-\gamma)\eta \sum_{k=1}^{N-2} (N-k-1)(r\eta)^k. \tag{A.6}$$

- A mixed strategy $\mathbf{x}_5 = (1-\mu)\mathbf{e}_2 + \mu\mathbf{e}_3 \leftrightarrow \mathbf{UD} \oplus \mathbf{DA}$, where (under appropriate conditions on the parameters) $0 < \mu < 1$ is the unique root of the equation

$$W(\mu) \equiv (1-\gamma)\mu \sum_{k=1}^{N-1} (r\mu)^k - \gamma = 0. \tag{A.7}$$

- Finally, there is a mixed strategy $\mathbf{y} = (\alpha, \beta, \chi)^{\mathrm{t}}$ such that $\chi = N\gamma/(N-1)/r$, and $\alpha, \beta > 0$. However, it is easy to show that $\mathbf{y}$ is not an ESS. Let us define $\mathbf{b} = (1-\chi, 0, \chi)^{\mathrm{t}}$. Now $\mathbf{b} \neq \mathbf{y}$, but $v(\mathbf{y} - \mathbf{b},\ \mathbf{y} + \varepsilon(\mathbf{b} - \mathbf{y})) = 0$ for any $\varepsilon > 0$.

Thus, it remains to test the evolutionary stability of the $\mathbf{x}_1$ through $\mathbf{x}_5$ potential ESS solutions.

It is easy to show that $\mathbf{x}_1$ is not an ESS:

$$\mathbf{x}_1 \neq \mathbf{e}_2 \quad \text{but } \xi_0(\mathbf{x}_1, \mathbf{e}_2) = -N\gamma. \tag{A.8}$$

To show that $\mathbf{x}_2$ is an ESS for all parameter values is equally easy:

$$\xi_0(\mathbf{x}_2, \mathbf{x}) = \gamma(Nx_1 + x_3). \tag{A.9}$$

The testing of $\mathbf{x}_3$ is somewhat more complicated:

$$\xi_0(\mathbf{x}_3, \mathbf{x}) = (1-r)F(r)x_1 + G(r)x_2, \tag{A.10a}$$

where

$$F(r) = (N-1) - (1-\gamma)\sum_{k=0}^{N-2} (N-k-1)r^k$$

and

$$G(r) = (1-\gamma)\sum_{k=1}^{N-1} r^k - \gamma.$$

(I) Since $0 < \gamma < 1$, $F(r)$ is monotone decreasing. Now, $F(0) = (N-1)\gamma > 0$, and $F(1) = (N-1)[N\gamma - (N-2)]/2 = N(N-1)(\gamma - \gamma_1)/2$. Consequently,
  (a) If $\gamma < \gamma_1$, then $F(1) < 0$. Thus, there exists a unique $0 < \rho_2 < 1$ such that $F(\rho_2) = 0$. That is, $F(r) > 0$ *iff* $r < \rho_2$.
  (b) If $\gamma_1 < \gamma$, then $F(1) > 0$. And therefore, $F(r) > 0$ for any $0 \leqslant r \leqslant 1$.

(II) $G(r)$ is monotone increasing, $G(0) = -\gamma < 0$, and $G(1) = (N-1) - N\gamma = N(\gamma_2 - \gamma)$. Consequently,
  (a) When $\gamma < \gamma_2$, $G(1) > 0$. Hence, there exists a unique $0 < \rho_1 < 1$ such that $G(\rho_1) = 0$. That is, $G(r) > 0$ *iff* $r > \rho_1$.
  (b) If $\gamma > \gamma_2$, then $G(r) < 0$ for $0 \leqslant r \leqslant 1$.

(III)

(a) $G(r)$ is monotone increasing and $G(\gamma) = -\gamma^{N-1} < 0 = G(\rho_1)$. Thus, $\gamma < \rho_1$.

(b) $F(r)$ is monotone decreasing and

$$F(\rho_1) = (1-\gamma)\sum_{k=1}^{N-1} k\rho_1^k > 0 = F(\rho_2).$$

Thus, $\rho_1 < \rho_2$.

(c) For $j = 1, 2$: let us denote the value of ρ_j at N by $\rho_j(N)$. Since, $F(\rho_2(N)) = 0$, we have

$$\frac{1}{1-\gamma} = \frac{1}{N-1}\sum_{k=0}^{N-2}(N-k-1)[\rho_2(N)]^k = \frac{1}{1-\rho_2(N)} - \frac{\rho_2(N) - [\rho_2(N)]^N}{[1-\rho_2(N)](N-1)}.$$

Now,

$$\frac{\rho_2(N) - [\rho_2(N)]^N}{[1-\rho_2(N)](N-1)} \xrightarrow[N\to\infty]{} 0.$$

Hence, $\rho_2(N) \xrightarrow[N\to\infty]{} \gamma$. Moreover, since $\gamma < \rho_1(N) < \rho_2(N)$ for any finite N, $\rho_1(N) \xrightarrow[N\to\infty]{} \gamma$.

(IV) Finally, $\mathbf{x}_3 \neq \mathbf{e}_1$, but $v(\mathbf{x}_3 - \mathbf{e}_1, \mathbf{x}_3 + \varepsilon(\mathbf{e}_1 - \mathbf{x}_3))|_{r=1} = 0$.

Hence, $\mathbf{x}_3 \leftrightarrow \mathbf{DA}$ is an ESS whenever

$$\gamma < \gamma_1 < \gamma_2 \quad \text{and} \quad \rho_1 < r < \rho_2$$
$$\text{or } \gamma_1 < \gamma < \gamma_2 \quad \text{and} \quad \rho_1 < r < 1. \tag{A.10b}$$

For $\mathbf{x}_4$, we see the following

$$\xi_0(\mathbf{x}_4, \mathbf{x}) = (N-1)r(\eta - \chi)x_2$$
$$\text{i.e. } \xi_0(\mathbf{x}_4, \mathbf{x}) = 0 \Leftrightarrow \mathbf{x} = \mathbf{x}_\lambda,$$
$$\text{where } \mathbf{x}_\lambda \equiv (1-\lambda)\mathbf{e}_1 + \lambda\mathbf{e}_3$$
$$\text{but } \xi_1(\mathbf{x}_4, \mathbf{x}_\lambda) = (1-r)H'(\eta)(\lambda - \eta)^2. \tag{A.11a}$$

Since $H(\eta)$ is monotone increasing, $\mathbf{x}_4$ is an ESS *only if* $\chi < \eta < 1$ and $r < 1$. However, $\chi < \eta < 1$ *iff* $H(\chi) < 0 < H(1)$. Now,

(I) $H(1) = -F(r) > 0$ *iff* $\gamma < \gamma_1$ and $r > \rho_2$.

(II) $H(\chi) = \chi[(1-\gamma)w(\gamma) - q(r)]$, where $q(r) = (N-1)^2 r/N$ and

$$w(\gamma) = \sum_{k=1}^{N-2}(N-k-1)[N_\gamma/(N-1)]^k.$$

Now, $F(r) < 0$ implies

$$\gamma < \frac{\sum_{k=1}^{N-1}(N-k-1)r^k}{\sum_{k=0}^{N-1}(N-k-1)r^k} < \frac{N-1}{N}r.$$

Thus, since $w(\gamma)$ and $q(r)$ are monotone increasing,

$$w(\gamma) < w\left(\frac{N-1}{N}r\right) \quad \text{and} \quad -q(r) < -q\left(\frac{N\gamma}{N-1}\right).$$

Hence,

$$H(\chi) < \chi\left[(1-\gamma)w\left(\frac{N-1}{N}r\right) - q\left(\frac{N\gamma}{N-1}\right)\right] = \chi F(r) < 0.$$

(III) Finally, $\mathbf{x}_4 \neq \mathbf{e}_1$, but $v(\mathbf{x}_4 - \mathbf{e}_1, \mathbf{x}_4 + \varepsilon(\mathbf{e}_1 - \mathbf{x}_4))|_{r=1} = 0$.

Hence, $\mathbf{x}_4 \leftrightarrow \mathbf{IA} \oplus \mathbf{DA}$ is an ESS whenever

$$N > 2, \gamma < \gamma_1 \quad \text{and} \quad \rho_2 < r < 1. \tag{A.11b}$$

We finish by demonstrating that $\mathbf{x}_5$ is not an ESS:

$$\mathbf{x}_5 \neq \mathbf{e}_2 \quad \text{but } \xi_0(\mathbf{x}_5, \mathbf{e}_2) = 0$$
and
$$\xi_1(\mathbf{x}_5, \mathbf{e}_2) = -\mu^2\left|\frac{\mathrm{d}W}{\mathrm{d}\mu}(\mu)\right|. \tag{A.12}$$

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